Name: _____ BME 105 Practice Final Exam Question

Student ID: _____

will have white patches

1. You are a cat breeder and have a female tortoiseshell cat (black and orange coat) that you would like to mate with a male black cat that is heterozygous for the piebald gene. What is the probability of getting a calico cat (black and orange coat with white patches)?

   As a reminder, coat color is determined by an X-linked gene. One allele causes orange pigment and the other allele causes black pigment. The piebald gene is an autosomal gene where the dominant allele, S, causes white patches. The piebald gene is epistatic to the coat color gene.

father

X^O – orange allele

X^B – black allele

S – white patches

s – no white patches

$X^O X^B s s \times X^B Y S s$

$P(X^O X^B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$

$P(S s) = \frac{1}{2}$

? $X^O X^B$ S –

$\frac{1}{4}$ $\frac{1}{2}$ = $\frac{1}{8}$

2. The following sequence of DNA from a eukaryotic genome contains a hypothetical, very short protein that is only 5 amino acids long.

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5'-AATAGCAGATGAGGTCTATCGATGGTCTTCTATGAGGTTCCCTCACTAACGCTTG-3'
3'-TTATCGTCTACTCCAGATAGCTACCAGAAGATACTCCAAGGGAGTGATTGCGAAC-5'
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a. What is the full amino acid sequence of the short polypeptide (you will be given the genetic code table for the exam)? As a reminder, splice site sequences start with GU and end with AG.

ATG AGG TTC CCT CAC TAA

Met Arg Phe Pro His

PAM NGG

last 6nt of sgRNA sequence is also ok.

b. You would like to use CRISPR/Cas9 to create a knockout of this polypeptide to investigate its function. This requires the design of an sgRNA. The crRNA portion of the sgRNA is typically 20nt long; however, for the purposes of this question, just indicate the first 6nt of the 20nt sgRNA sequence. Give one possible 6nt sequence that would be used to potentially knockout this gene?

Best NGG sites highlighted in green

OK NGG sites highlighted in yellow → will cut in intron, no near splice sites nor in/near coding sequence

Possible sequences:

5'-CAGAUG-3'

5'-UAGUGA-3'

5'-UAGCAG-3'

or

5'-UAUCGA-3'

5'-UCUAUC-3'

or

5'-TCTATG-3'

Sequence is cut off, so these are the last 6nt of the 20nt crRNA sequence.